

# Skills Summary

## Placing Fractions on a Number Line

*Use this reference while completing your worksheet or when studying.*

**The two counts.** On a number line, a fraction is a *distance from 0*, not a picture of a pizza.

**Bottom number (denominator):** how many equal parts one whole was cut into. Count the *parts*, never the tick marks.

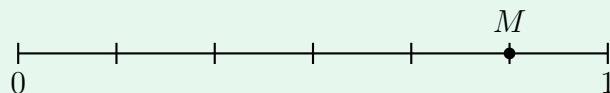
**Top number (numerator):** how many of those parts you travel, starting at 0.

So  $\frac{3}{4}$  means: cut the whole into 4 equal parts, then go 3 of them.

### 1. Name the point that is already marked

Read the labels at the ends first — they tell you where one whole finishes. Then count the equal parts between 0 and 1 to get the bottom number, and count parts from 0 to the dot to get the top number.

**Example:** Name point  $M$ .



**Step 1.** The dot is already placed, so the job is to read it — count parts for the bottom number, then count travel for the top number.

**Step 2.** Equal parts from 0 to 1: six, so the parts are sixths. Parts travelled from 0 to  $M$ : five. Five parts of a whole cut into sixths:  $M = \frac{5}{6}$

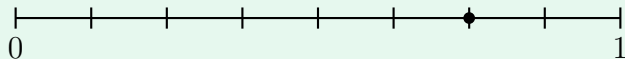
**Try it:** A line from 0 to 1 is cut into 3 equal parts. A dot sits on the first tick after 0. What fraction names the dot?

check:  $\frac{1}{3}$

## 2. Place a fraction you are given

Work from the bottom number outward: it tells you how many equal parts the whole needs. Then count that many parts from 0 — the top number of them — and put the dot there. Two fractions that land on the same point name the same number, so  $\frac{2}{4}$  and  $\frac{1}{2}$  are the same place.

**Example:** Mark  $\frac{6}{8}$ , then name the point with a simpler fraction.



**Step 1.** The bottom number 8 says the whole is cut into eighths, so count six eighths from 0 and mark the sixth tick.

**Step 2.** Now group the eighths in twos: two eighths make one fourth, so the eight parts become four fourths and the dot lands on the third of them. The same point, named in bigger parts:  $\frac{3}{4}$

**Try it:** On a line cut into 6 equal parts, mark  $\frac{2}{6}$ . What simpler fraction names that same point?

check:  $\frac{2}{6}$

## 3. Compare two fractions on the line

Whichever fraction lands *farther from 0* is the greater one — that is all comparing means on a number line.

$>$  means *is greater than*       $<$  means *is less than*

When the top numbers match, the fraction with the *smaller* bottom number wins, because fewer cuts make longer parts.

**Example:** Which is greater,  $\frac{3}{4}$  or  $\frac{3}{6}$ ?

**Step 1.** Both travel three parts, so the winner is decided by which parts are longer — compare the sizes of the cuts, not the numbers.

**Step 2.** Cutting a whole into 4 makes longer parts than cutting it into 6, so three of them reach farther. On a line,  $\frac{3}{4}$  sits past the middle while  $\frac{3}{6}$  sits exactly on it.  $\frac{3}{4}$  is farther from 0, so the sign is  $>$

**Try it:** Put  $\frac{1}{2}$  and  $\frac{5}{8}$  on one eighths line (remember  $\frac{1}{2}$  is four eighths). Which sign makes  $\frac{1}{2}$   $\frac{5}{8}$  true?

check:  $>$

**(!) Watch out:** count the *spaces* between ticks, not the ticks themselves — a line cut into fourths has five tick marks. And check the label at the right-hand end before you start: if it says 2, the parts between 0 and 1 are still the unit, so a dot past the middle can be more than one whole.